SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

Time Duration : 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Code of Conduct:

I M . Indupriya (192372085)) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
indupriya
Signature of the student:

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive learning approach

This is a supervised inductive learning problem: you have labeled historical patient records (diabetic/non-diabetic outcome known) and want the model to infer a general classification rule that generalizes to new, unseen patients. A decision-tree/rule-induction approach fits naturally here because clinicians value interpretable "if-then" rules over opaque scoring.

- **Target variable:** Diabetes Outcome (binary: 1 = diabetic, 0 = non-diabetic).
- **Key features and justification:**
  - **Glucose level** — the direct physiological marker used in WHO/ADA diagnostic criteria; strongest single predictor.
  - **BMI** — proxy for insulin resistance, a core mechanism in Type 2 diabetes.
  - **Age** — insulin sensitivity declines with age, and prevalence rises sharply after ~45.
  - **Family history** — captures genetic predisposition not visible in current lab values.

(b) Handling class imbalance

Assume the dataset skews ~65% non-diabetic / 35% diabetic. Recommended strategy: SMOTE (Synthetic Minority Oversampling Technique) combined with class weighting.

- SMOTE generates synthetic minority-class (diabetic) samples by interpolating between existing minority neighbors, rather than duplicating rows — this avoids the overfitting risk of naive oversampling and the information loss of under-sampling the (clinically valuable) majority class.
- Class weights (e.g., class_weight='balanced') additionally penalize the model more heavily for misclassifying diabetic cases in the loss function — cheap to apply and works well alongside SMOTE.
- Justification: in a medical screening context, a false negative (missed diabetic patient) is far costlier than a false positive, so the balancing strategy should specifically push the model toward higher recall on the minority class.

(c) Data split and preprocessing

- Split: 70:30 (train:test), stratified on the outcome variable so both sets preserve the same diabetic:non-diabetic ratio.
- Justification: medical datasets are often modest in size (hundreds, not millions, of records), so 70% preserves enough training signal for the model to learn reliable patterns, while a 30% held-out test set (not a smaller 10–20%) gives a large enough sample for stable, trustworthy precision/recall estimates — critical when a wrong metric could mislead a clinical deployment decision.
- **Preprocessing steps:**
  - **Missing/implausible value handling** — e.g., glucose or BMI recorded as 0 is biologically implausible; impute using median/mean grouped by outcome class.
  - **Normalization/standardization** (Min-Max or Z-score) — features like age (years) and glucose (mg/dL) are on very different scales; this matters for distance- or gradient-based models (logistic regression, k-NN) even though tree models are scale-invariant.
  - **Encoding** — binary encode family history (Yes/No → 1/0); one-hot encode any multi-category fields.
  - **Impact:** improves convergence speed and coefficient comparability for statistical models, prevents any single large-scale feature from dominating distance calculations, and ensures fair feature-importance comparison across models.

Task 2: Decision Tree for Diagnosis

(a) Building the tree — root node justification

Training set (70% split), n = 538: 350 non-diabetic, 188 diabetic.

$$
Gini(root)= 1 - \left[\left(\frac{350}{538}\right)^2 + \left(\frac{188}{538}\right)^2\right] = 1 - (0.4225+0.1218) = 0.4557
$$

Testing a split on **Glucose ≤ 127**:

Branch	n	Diabetic	Non-diabetic	Gini
Glucose ≤ 127	343	76	267	0.345
Glucose > 127	195	112	83	0.489

$$
Gini_{split} = \frac{343}{538}(0.345) + \frac{195}{538}(0.489) = 0.397
$$

$$
InfoGain_{Glucose} = 0.4557 - 0.397 = 0.059
$$

Repeating this for BMI (gain ≈ 0.031) and Age (gain ≈ 0.021) shows Glucose yields the highest Gini gain, so Glucose is selected as the root node.

First three levels (illustrative):

```
graph TD
  A["Root: Glucose ≤ 127?"] -->|Yes| B["BMI ≤ 30?"]
  A -->|No| C["Age ≤ 45?"]
  B -->|Yes| D["Healthy - Low risk"]
  B -->|No| E["Family History = Yes?"]
  C -->|Yes| F["BP ≤ 80?"]
  C -->|No| G["Diabetic - High risk"]
```

(b) Model evaluation

Test set (n = 230; 80 actual diabetic, 150 actual non-diabetic):

	Predicted Diabetic	Predicted Non-diabetic
Actual Diabetic	TP = 58	FN = 22
Actual Non-diabetic	FP = 18	TN = 132

$$
Accuracy = \frac{58+132}{230} = 82.6\%,\quad Precision = \frac{58}{76} = 76.3\%
$$

$$
Recall = \frac{58}{80} = 72.5\%,\quad F1 = 2\cdot\frac{0.763 \times 0.725}{0.763+0.725} = 74.4\%
$$

False Negatives (22 patients): these are undiagnosed diabetics who leave the clinic without treatment — clinically the most dangerous error, since untreated diabetes leads to cardiovascular, renal, and neurological complications. To reduce FN:

- Lower the classification probability threshold (trade some precision for higher recall/sensitivity).
- Apply cost-sensitive learning that penalizes FN more heavily than FP in the splitting criterion.
- Use the model as a screening trigger (flag borderline cases for confirmatory HbA1c testing) rather than a sole diagnostic gate.

(c) Pre-pruning vs. post-pruning

Model	Train Accuracy	Test Accuracy	Test Recall
Unpruned (full-depth)	~98%	76.0%	68%
Pre-pruned (max_depth=4)	85%	80.4%	70%
Post-pruned(cost-complexity, ccp_alpha=0.015)	88%	83.9%	75.6%

Post-pruning grows the full tree first, then removes branches whose contribution to accuracy doesn't justify their added complexity (measured by the cost-complexity parameter α). This generally generalizes better than pre-pruning because pruning decisions are made with full knowledge of the tree structure, rather than stopping growth early based on local heuristics.

For clinical deployment, the post-pruned tree is preferable — it retains the clinically meaningful splits (Glucose, BMI) while achieving both higher overall accuracy and higher recall (fewer missed diabetes cases), and remains interpretable enough for clinicians to trust the decision path.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs. Decision Tree

Model	**Accuracy**	**AUC-ROC**
Logistic Regression	80.0%	0.84
Post-pruned Decision Tree	83.9%	0.81

Logistic Regression yields better-calibrated probability estimates (higher AUC-ROC), which matters for risk stratification — grouping patients into low/medium/high-risk bands rather than a hard binary call. It's preferable when:

- Feature relationships are roughly linear/log-linear in the log-odds.
- The dataset is smaller, and a model with fewer parameters generalizes more robustly.
- Interpretable odds ratios are needed for clinical communication (e.g., "each 1-unit BMI increase raises diabetes odds by X%").

The Decision Tree remains preferable when non-linear feature interactions matter or when frontline staff need a transparent rule-based decision path rather than a probability score.

(b) Feature importance comparison

Rank	**DecisionTree(Gini importance)**	**Logistic Regression (odds ratio)**
1	Glucose (0.42)	Glucose (OR ≈ 2.3)
2	BMI (0.24)	BMI (OR ≈ 1.8)
3	Age (0.15)	Family History (OR ≈ 1.6)

Both models agree that Glucose and BMI are the top two predictors — consistent with clinical literature, since glucose is the diagnostic marker itself and BMI reflects the insulin-resistance mechanism. They diverge on the third factor: the tree favors Age (capturing a non-linear risk jump past a threshold, e.g., age > 45), while logistic regression favors Family History (a fairly consistent proportional risk multiplier across all ages, which linear log-odds models capture well). This suggests age acts as a threshold effect while family history acts as a steady multiplicative risk factor.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP formulation

- **States (S):** S1 = Healthy/Low-risk, S2 = Pre-diabetic/Moderate-risk, S3 = Diabetic-controlled, S4 = Diabetic-uncontrolled/High-risk. *Justification:* discretizing continuous glucose/HbA1c/BMI trends into clinically recognized stages (aligned with ADA staging) keeps the state space tractable while remaining medically meaningful.
- **Actions (A):** {Diet, Exercise, Medication, Monitor}. *Justification:* mirrors the real intervention menu available to physicians, ranging from low-risk lifestyle actions to higher-impact but higher-risk pharmacological ones.
- **Reward function:** positive reward for transitioning to a lower-risk state (e.g., +10), penalty for stagnation/worsening (−5 / −20), and a small cost penalty on Medication to reflect side-effect/cost trade-offs — this discourages unnecessary escalation when a lighter intervention would suffice.
- **Transition dynamics:** $P(s'|s,a)$ estimated empirically from historical patient trajectories (e.g., what fraction of pre-diabetic patients moved to "Healthy" after 3 months on Diet+Exercise). *Justification:* patient response to treatment is inherently stochastic (adherence, genetics, unmeasured confounders), so a probabilistic — not deterministic — transition model is required.

(b) Q-Learning table updates

Update rule ($\alpha = 0.1$, $\gamma = 0.9$):

$$
Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a'}Q(s',a') - Q(s,a)\right]
$$

Iteration 1 (episodes 1–3, Q initialized at 0):

Episode	(State, Action)	Reward	Next State	Updated Q
1	(S2, Diet)	+8	S1	$0 + 0.1[8 + 0 - 0] =$ **0.80**
2	(S3, Medication)	+5	S3	$0 + 0.1[5 + 0 - 0] =$ **0.50**
3	(S4, Medication)	−2	S3	$0 + 0.1[-2 + 0.9(0.5) - 0] =$ **−0.155**

Iteration 2 (episodes 4–5, revisiting states with updated Q-values):

Episode	(State, Action)	Reward	Next State	Updated Q
4	(S2, Diet)	+9	S1	$0.80 + 0.1[9 + 0 - 0.80] =$ **1.62**
5	(S3, Medication)	+6	S3	$0.50 + 0.1[6 + 0.9(0.5) - 0.50] =$ **1.10**

Convergence criterion: train until the maximum change in any Q-value between successive full sweeps falls below a small threshold (e.g., $\Delta < 0.01$), or a fixed episode budget (e.g., 1,000+ episodes) is reached — whichever comes first — with a decaying learning rate α to satisfy standard stochastic-approximation convergence conditions.

(c) Final policy and ethical considerations

Learned policy: $$\pi(s) = \arg\max_a Q(s,a)$$ — e.g., π(S1)=Monitor, π(S2)=Diet, π(S3)=Medication, π(S4)=Medication (with Exercise as a close second in S3, worth clinical review).

Ethical considerations for deployment:

- **Patient safety:** the agent should function as a *decision-support* recommender, not an autonomous prescriber — action recommendations require clinician sign-off, especially for Medication, since live exploration of a "wrong" action on real patients (as standard RL exploration would demand) is not ethically acceptable. Training should rely on offline/batch RL from historical trajectories or a simulated patient model.

- **Bias:** if the training data underrepresents certain age groups, ethnicities, or socioeconomic backgrounds, the learned policy may not generalize equitably — fairness audits across subgroups are essential before deployment.
- **Explainability:** clinicians and patients need to understand *why* a recommendation was made (e.g., via policy visualization or feature-attribution methods), both for clinical trust and to satisfy regulatory expectations (FDA/CE clinical decision-support requirements, GDPR-style "right to explanation").
- **Accountability:** maintain human-in-the-loop oversight with audit trails comparing AI recommendations to actual clinician decisions, so responsibility for treatment outcomes remains clearly assignable.